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MADRAS INSTITUTE OF TECHNOLOGY, CHROME PET

Department of Electronics Engineering

B.E. Electronics and Communication Engineering

EC23501 ANTENNA AND WAVE PROPAGATION

Semester : 05

Simulation Report

DESIGN OF CONFORMAL ANTENNA

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INDEX

S.No.	Content	Page No.
1	Aim	3
2	Software Used	3
3	Theory	3
4	Design Section	4
5	Design and Measured Values	
	1. Planar structure	6
	2. Cylindrical with radius $R = 9\text{ mm}$	7
	3. Cylindrical with radius $R = 17\text{ mm}$	8
	4. Cylindrical with radius $R = 30\text{ mm}$	9
6	Inference	11
7	Conclusion	11
8	Reference	12

DESIGN OF CONFORMAL ANTENNA

AIM

To design and analyze a **4×4 Conformal microstrip patch antenna array** operating at **50 GHz**, and to study how **bending radius** affects the antenna's **resonant frequency, bandwidth, return loss, VSWR, gain, and directivity** when the structure is bent conformally over cylindrical surfaces of different radii.

SOFTWARE USED

CST Microwave Studio Suite 2025

THEORY

4×4 Conformal Microstrip Patch Antenna Array

A 4×4 conformal microstrip patch antenna array consists of sixteen rectangular microstrip patches arranged in a 4×4 configuration on a flexible dielectric substrate. Designed to operate around 50 GHz, it combines the high gain and directivity of a planar array with the mechanical flexibility of a conformal surface. When the array is bent over cylindrical structures, slight changes occur in resonant frequency and radiation pattern due to variations in the effective dielectric constant and surface current path. However, the antenna maintains good impedance matching, stable bandwidth, and high efficiency, making it ideal for compact and high-performance millimeter-wave systems.

Advantages

- High gain and directivity due to array configuration
- Lightweight and flexible for surface-mounted applications
- Stable performance even under bending conditions
- Compact structure suitable for integration into devices
- Capable of wide angular coverage for wearable systems

Applications

- Millimeter-wave 5G/6G communication systems
- Wearable and body-centric wireless devices
- Radar and imaging systems
- Aerospace and automotive surface antennas
- High-speed short-range data links (50–60 GHz band)

DESIGN SECTION

Frequency of operation: 50 GHz

Antenna type: 4x4 microstrip patch array

Substrate: Polytetrafluoroethylene 20 mm x 25 mm

- Dielectric constant (ϵ_r) = 2.2
- Thickness = 0.127 mm

Width of the Patch (W)

$$W = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}}$$

where:

- W = Width of the patch (m)
- c = Velocity of light (3×10^8 m/s)
- f_r = Resonant frequency (Hz)
- ϵ_r = Relative permittivity of substrate

For PTFE ($\epsilon_r = 2.2$) and $f_r = 50$ GHz, $W = 2.3717$ mm

Effective Dielectric Constant (ϵ_{eff})

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W}\right)^{-\frac{1}{2}}$$

where:

- h = Substrate thickness (m)

For $\epsilon_r = 2.2$, $h = 0.127$ mm, $W = 1.65$ mm, $\epsilon_{eff} = 2.06815$

Effective Length (Leff)

$$L_{eff} = \frac{c}{2f_r \sqrt{\epsilon_{eff}}}$$

For $f_r = 50$ GHz, $\epsilon_{eff} = 1.6$, $L_{eff} = 2.08608$ mm

Length Extension due to Fringing (ΔL)

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{eff} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{eff} - 0.258) \left(\frac{W}{h} + 0.8 \right)}$$

For $\epsilon_{eff} = 1.6$, $W/h = 13$, $h = 0.127$ mm, $\Delta L = 0.06847$ mm

Actual Physical Length (L)

$$L = L_{eff} - 2\Delta L$$

Using $L_{eff} = 2.08608$ mm, $\Delta L = 0.06847$ mm, $L = 1.94913$ mm

Patch dimensions: 2.3717 mm $\times$ 1.94913 mm, thickness = 0.018 mm

Feed lines: Width = 0.12 mm

Configurations tested:

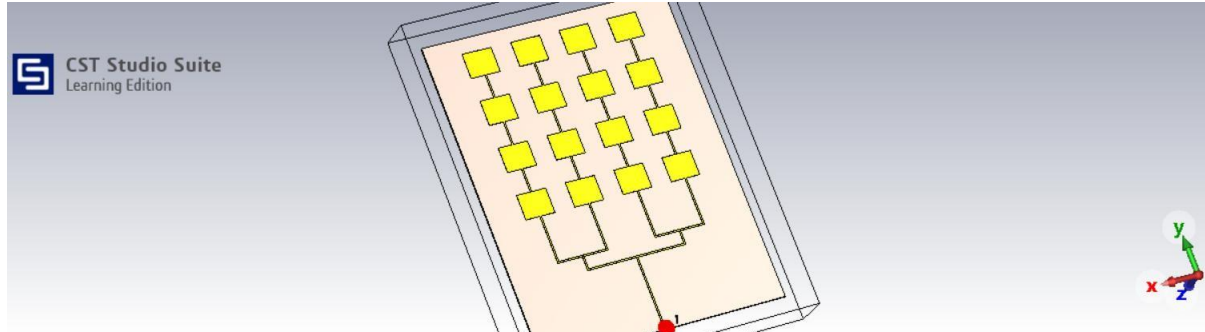
5. Planar structure
6. Cylindrical with radius $R = 9$ mm
7. Cylindrical with radius $R = 17$ mm
8. Cylindrical with radius $R = 30$ mm

Parameter List		
Name	Expression	Value
antz	= 0.018	0.018
feedx	= 0.12	0.12
feedxsmall	= 0.12	0.12
subz	= 0.127	0.127
d	= 1.62	1.62
feedy	= 1.83	1.83
anty	= 1.94913	1.94913
antx	= 2.3717	2.3717
feedxmax	= 3.28	3.28
feedymax	= 3.735	3.735
subx	= 20	20
suby	= 25	25

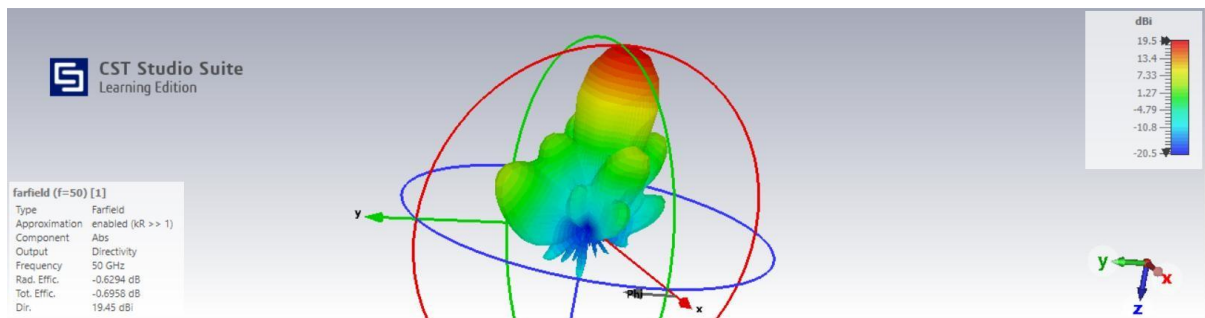
DESIGN AND MEASURED RESULTS

Planar Structure

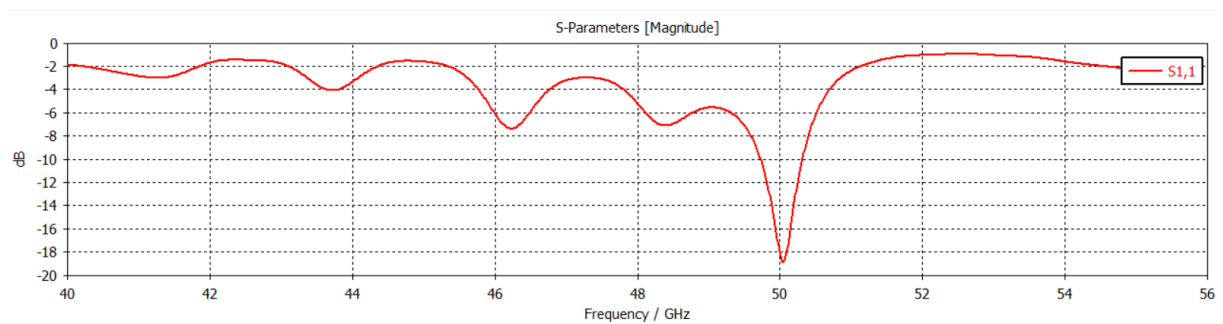
Antenna Design



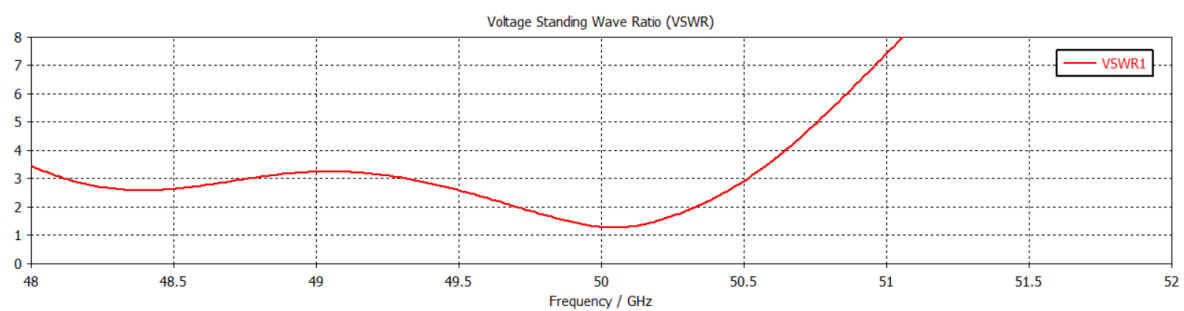
Radiation Pattern



S-Parameter

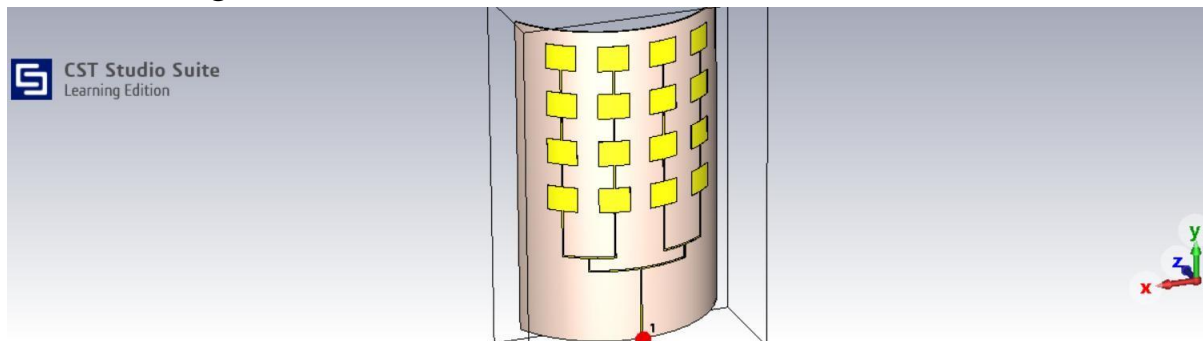


VSWR

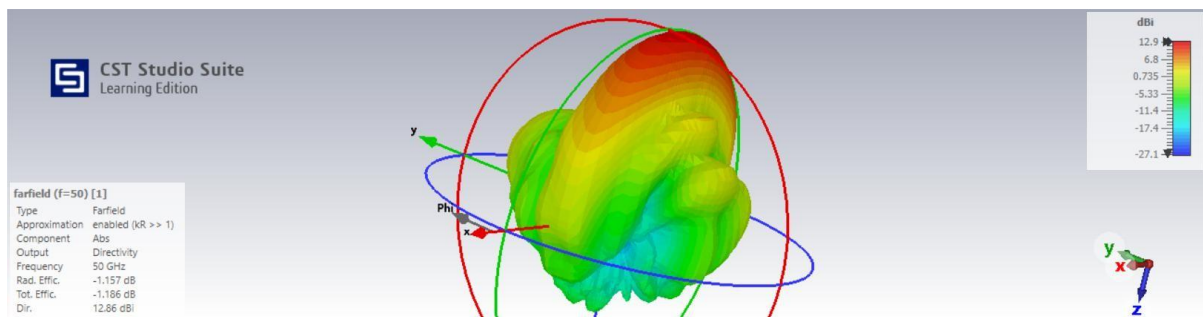


Cylindrical with radius $R = 9$ mm

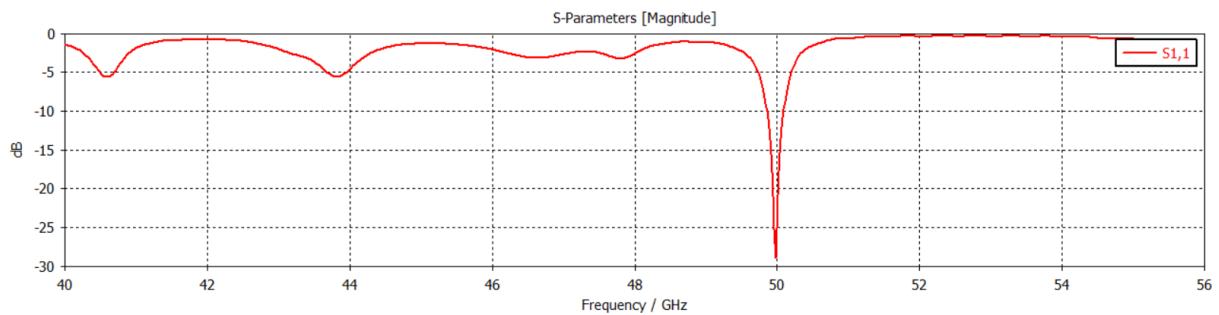
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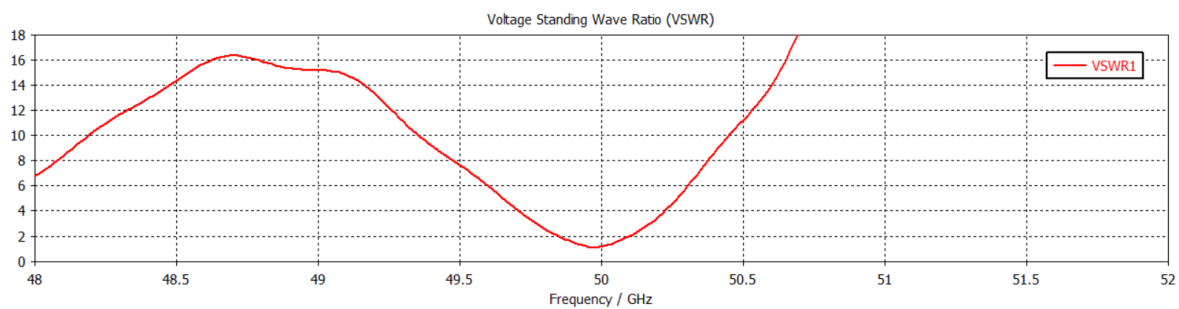
Radiation Pattern



S-Parameter

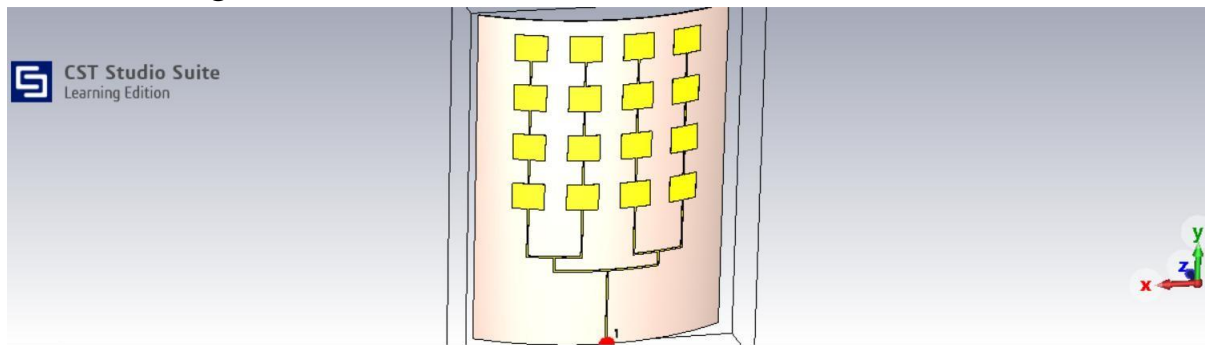


VSWR

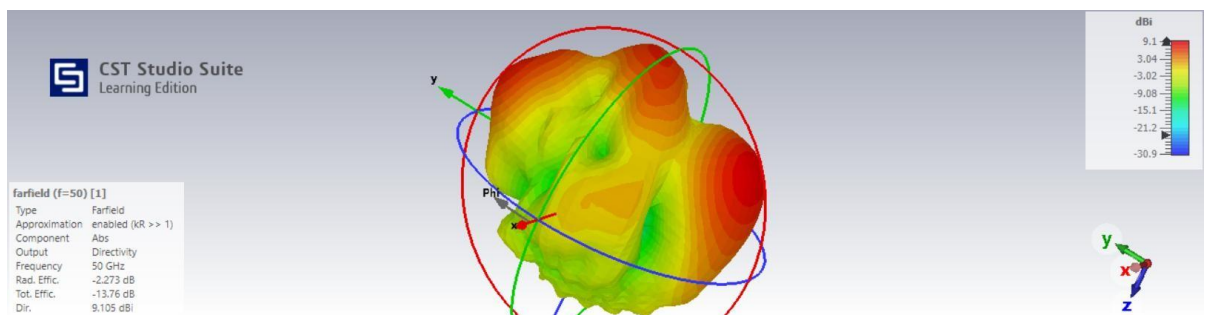


Cylindrical with radius $R = 17$ mm

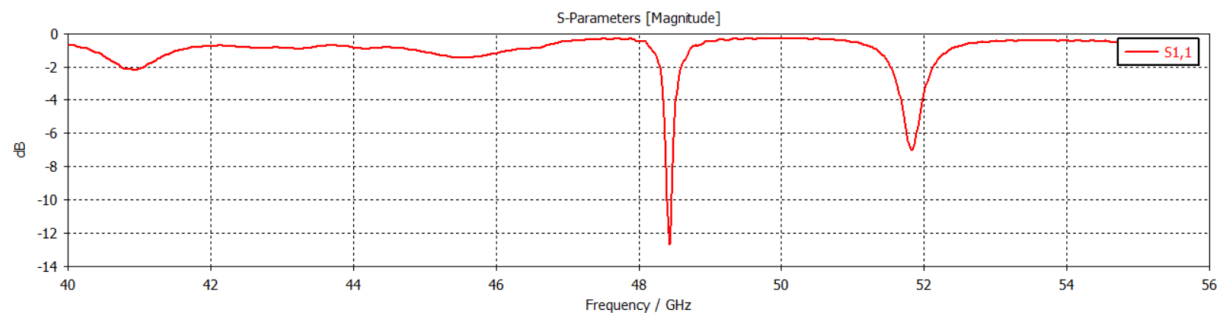
Antenna Design



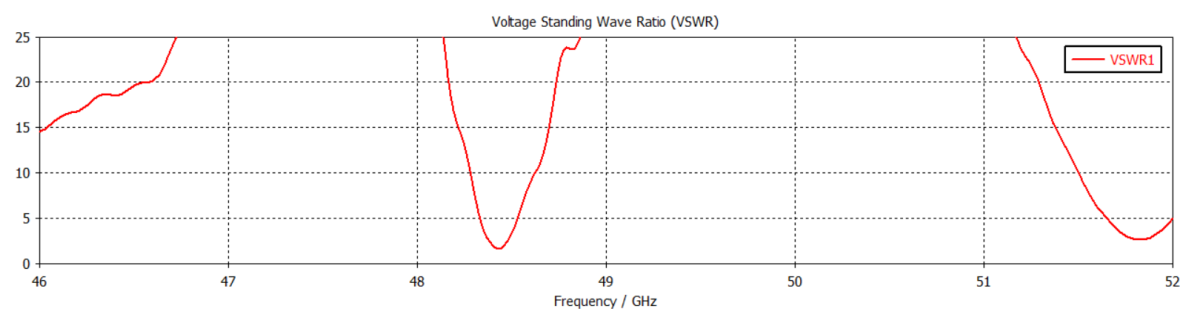
Radiation Pattern



S-Parameter

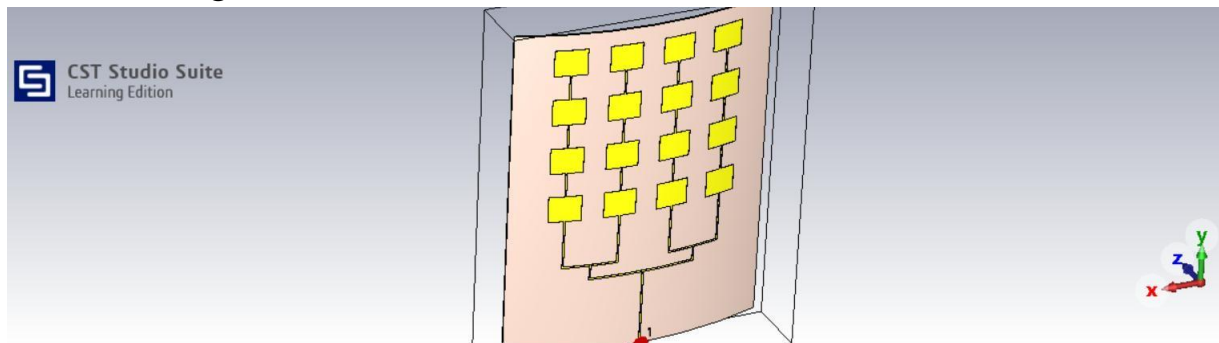


VSWR

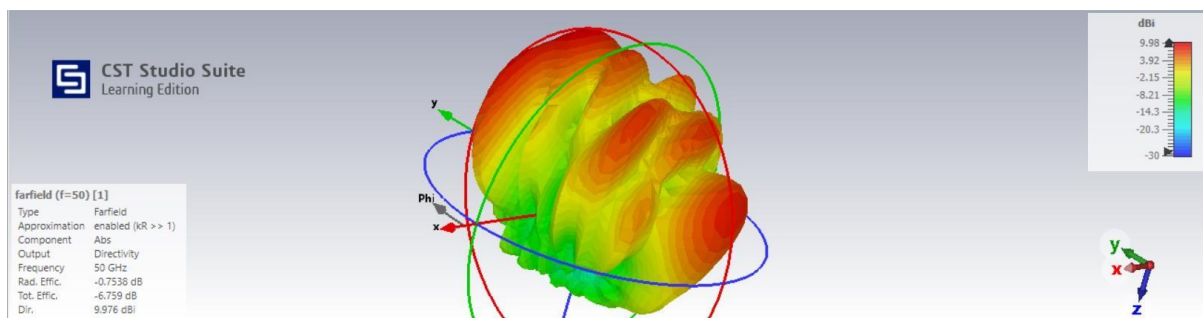


Cylindrical with radius $R = 30$ mm

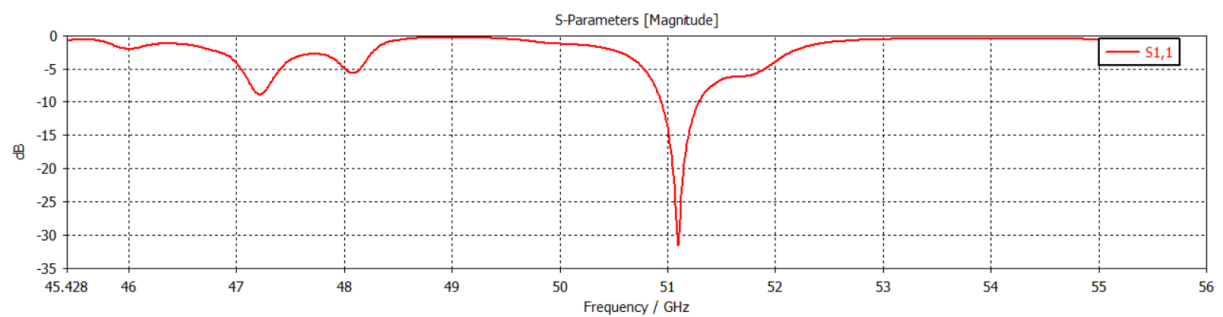
Antenna Design



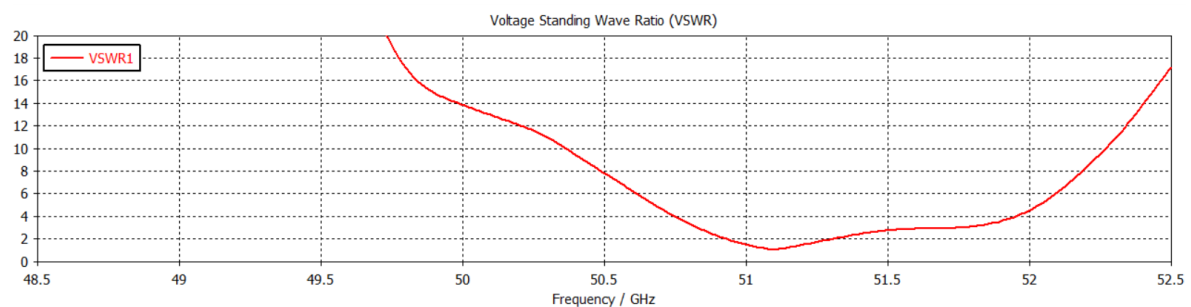
Radiation Pattern



S-Parameter



VSWR



Measured Values

Configuration	Radius (mm)	Resonant Frequency (GHz)	Bandwidth (MHz)	Return Loss (dB)	VSWR	Gain (dBi)	Directivity (dBi)
Planar	∞	50.05011	587.75	-18.86668	1.281145	18.82	19.45
Conformal (R = 9 mm)	9	49.975	236.07	-28.88288	1.074625	11.7	12.86
Conformal (R = 17 mm)	17	48.43	57.86	-12.72839	1.6	6.832	9.105
Conformal (R = 30 mm)	30	51.1	351.22	-31.60766	1.055006	9.222	9.976

INFERENCE

- The planar array resonates at 50 GHz, while bending changes the frequency slightly — 48.43 GHz ($R = 17$ mm) and 51.1 GHz ($R = 30$ mm).
- Return loss stays below -12.8 dB and VSWR between 1.1–1.6, showing good impedance matching.
- Bandwidth remains stable around 58-590 MHz for all cases.
- Gain varies from 6.832 dB to 18.82 dB, reducing slightly with tighter curvature.
- Directivity ranges between 9.1–19.45 dB, and the radiation beam becomes broader when bent.
- Overall, the antenna maintains excellent performance and stability, proving suitable for flexible and wearable 50 GHz millimeter-wave applications.

CONCLUSION

The **4×4 conformal microstrip patch antenna array** operating at **50 GHz** was successfully designed and analyzed. The study showed that bending the antenna over different radii slightly affects its resonant frequency, gain, and radiation pattern, while maintaining stable impedance matching and bandwidth. The planar and conformal configurations achieved good return loss (below -12.8 dB), low VSWR, and high gain (around 18.82 dB). These results confirm that the antenna provides high performance, flexibility, and reliability even under curvature, making it highly suitable for millimeter-wave, wearable, and surface-mounted communication systems.

REFERENCE

Semkin V., Bisognin A., Kyrö M., Kolmonen V-M., Luxey C., Ferrero F., Devillers F., Räisänen A.V. “*Conformal Antenna Array for Millimeter-Wave Communications: Performance Evaluation,*”
International Journal of Microwave and Wireless Technologies, Feb 2015.